

Erik Rasmussen

Software Engineer | Des Moines, IA

📞 (515) 782-2960

✉ erik.rasmussen@unmango.dev

🌐 [linkedin.com/in/unstoppablemango](https://www.linkedin.com/in/unstoppablemango)

🐙 github.com/UnstoppableMango

Summary

Open-source enthusiast from Des Moines with a decade of experience developing cloud-native solutions and line-of-business enterprise applications. Hobby sysadmin with a passion for functional programming, software architecture, and language design.

Stack

- **Languages:** Go, TypeScript/JavaScript, C#, F#, Rust, Haskell, C, C++, Java, Nix, Bash, PowerShell
- **Technologies:** Kubernetes, Terraform, Pulumi, Docker, Helm, Flux, GitHub Actions, Azure DevOps, gRPC, GraphQL, React (Native), Angular, Azure, AWS, SQL Server, PostgreSQL, Ceph
- **Certifications:** Certified Kubernetes Application Developer (CKAD), Certified Kubernetes Administrator (CKA)
- **Interests:** DevOps, Linux, compilers, language design, code generation, polyglot, event-driven architecture

Experience

Source Allies

Team Lead and Principal Consultant

*Urbandale, IA
Aug. 2020 - Present*

- Led teams of developers in an Agile environment to deliver high-quality software solutions.
- Mentored junior developers and conducted code reviews to maintain code quality and best practices.
- Actively engaged with both technical and non-technical clients to gather requirements, provide guidance, and ensure successful project delivery.
- Developed custom Kubernetes operators using Go to automate application-team infrastructure provisioning.
- Administered 60+ node Kubernetes clusters spanning from on-premise Nutanix and VMware hardware to AWS virtual machines running BottleRocket.
- Maintained efficient and re-usable CI/CD pipelines for application delivery teams using GitHub Actions Runner Controller.
- Architected large scale data pipelines in Azure supporting genome comparisons and other agricultural research with Azure Data Factory and Synapse.
- Migrated legacy Windows line-of-business application workloads to cloud-native Azure solutions.
- Built a rewards platform in Azure with ASP.NET and GraphQL supported by React Native mobile and web applications.
- Designed event-driven microservice architectures spanning multiple domains to enable real-time application updates and eventual consistency.
- Analyzed iOS and Android application performance and introduced automated benchmarking into the development lifecycle.
- Co-operated with design teams to implement user-friendly and accessible web and mobile application interfaces built on design tokens for re-usability.

Kum & Go

Application Developer

*Des Moines, IA
Jan. 2018 - Aug. 2020*

- Developed a food ordering management solution to support store associates completing orders submitted via mobile and web applications.
- Built automated CI/CD pipelines on Azure Pipelines for application deployment.
- Designed and implemented the service architecture for a cloud based enterprise food menu management solution in Azure.
- Migrated legacy .NET applications to cloud-native ASP.NET and .NET Core on Azure.
- Introduced new standards for automated testing and code quality analysis based on industry best practices.
- Collaborated with cross-functional teams to gather requirements and deliver high-quality software solutions.

Principal Financial Group

Application Developer Intern

*Des Moines, IA
Jun. 2017 - Aug. 2017*

- Performed application feature development supporting the insurance underwriting process.
- Introduced new integration testing practices for Java REST APIs using mock REST clients and re-usable Docker test environments.
- Refactored and re-structured legacy insurance policy applications to improve overall system performance.
- Deployed automated CI/CD pipelines for Java Spring web services on WebSphere using Jenkins.
- Developed testing suites with Gherkin and Cucumber to support behavior-driven development and engage non-technical team members.

Kum & Go

Application Developer Intern

*West Des Moines, IA
Jan. 2016 - May 2017, Sep. 2017 - Dec. 2017*

- Implemented a customer data management tool for the Customer Relations team built on AngularJS and backed by an ASP.NET REST API.
- Automated processes for updating and managing customer rewards program information.

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Side Projects

All projects are hosted on GitHub

- **unmango/go-make**: Makefile abstract syntax tree and parser written in Go.
- **pulumiverse/pulumi-talos**: A Pulumi provider for Talos cluster node provisioning.
- **unmango/pulumi-baremetal**: A Pulumi provider for baremetal host provisioning using a gRPC client/server model.
- **the-cluster**: Homelab kubernetes infrastructure built on Pulumi and Flux.
- **unmango/cloudflare-operator**: A kubernetes operator and ingress controller for Cloudflare resources.
- **advent-of-code**: Advent of Code solutions in various languages including Haskell, F# and Rust.
- **CliWrap.FSharp**: F# bindings for the C# library CliWrap.
- **dotfiles**: Personal dotfiles and NixOS configuration.

Education

University of Wisconsin Platteville
B.S. IN SOFTWARE ENGINEERING

Platteville, Wisconsin
Sep. 2013 - Dec. 2017

- Dean's Honours List